STT 861: Theory of Probability & Statistics I

Chapter 4.1–4.2: Joint and Marginal Distributions, Conditional Distributions and Independence

Guanqun Cao

Department of Statistics & Probability
Michigan State University

Nov 10, 2026 — Wells Hall C506

Reference: Casella & Berger Ch. 4, Sections 4.1 and 4.2

Where we are: from one variable to several

So far (Chapters 1–3) every model described **one** random variable. Real data record several quantities on each unit.

Today

- **§4.1** Joint distributions (pmf, pdf, cdf) and the **marginals** they contain.
- **§4.2 Conditional** distributions, conditional mean and variance, and **independence**.

Roadmap of Chapter 4

- 4.1 Joint & marginal ← today
- 4.2 Conditional & independence ← today
- 4.3 Bivariate transformations
- 4.4 Hierarchical & mixture models
- 4.5 Covariance & correlation
- 4.6–4.7 Multivariate distributions; inequalities

Motivation: why a single variable is not enough

Situation	**Pair** (X, Y)
A patient's height and weight	body measurements
Two dice: the larger and the smaller value	$(\max, \min)$
A stock's return today and tomorrow	time-dependent pair
Number of claims and the size of a claim	frequency and severity
Lifetime of component 1 and of component 2	reliability of a system
A random point in a plane	(X, Y) coordinates

Think (1 min)

Knowing the distribution of X alone and the distribution of Y alone: can you recover the distribution of the pair (X, Y)? *(No! We will see two different joint distributions with identical marginals.)*

Random vectors and joint pmf

Definition

An n-dimensional **random vector** is a function from the sample space S into $\mathbb{R}^n$. For $n = 2$ we write (X, Y). It is **discrete** if its range is countable.

Joint pmf

For a discrete random vector the **joint pmf** is

$$f_{X,Y}(x, y) = P(X = x,\ Y = y), \qquad (x, y) \in \mathbb{R}^2.$$

It satisfies $f \geq 0$ and $\sum_x \sum_y f(x, y) = 1$. For any $A \subset \mathbb{R}^2$:

$$P\big((X, Y) \in A\big) = \sum_{(x,y)\in A} f(x, y).$$

Expectation of a function of (X, Y)

$$\mathbb{E}\, g(X, Y) = \sum_x \sum_y g(x, y)\, f(x, y) \quad \Big(\text{continuous case: } \iint g(x, y) f(x, y)\, dx\, dy\Big).$$

Example: three coin tosses

Setup

Toss a fair coin 3 times. Let $X = 1$ if the first toss is heads (0 otherwise) and $Y =$ total number of heads.

	$y=0$	$y=1$	$y=2$	$y=3$	$f_X(x)$
$x=0$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0	$\frac{1}{2}$
$x=1$	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{1}{2}$
$f_Y(y)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1

Reading the table

- Interior = joint pmf; row/column sums = marginals.
- $P(X=1, Y=2) = \frac{2}{8}$: first toss H and exactly one H in the last two.
- $P(X=1, Y=0) = 0$: the first toss is H, so $Y \geq 1$.

$\mathbb{E}(XY)$

$$\mathbb{E}(XY) = 1 \cdot \left(1 \cdot \tfrac{1}{8} + 2 \cdot \tfrac{2}{8} + 3 \cdot \tfrac{1}{8}\right) = 1.$$

Marginal distributions

Theorem (marginal pmf)

Let (X, Y) be discrete with joint pmf $f_{X,Y}$. Then

$$f_X(x) = P(X = x) = \sum_y f_{X,Y}(x, y), \qquad f_Y(y) = \sum_x f_{X,Y}(x, y).$$

Why

$\{X = x\}$ is the disjoint union over y of the events $\{X = x, Y = y\}$, so add their probabilities.

From the coin table

f_Y is Binomial$(3, \frac{1}{2})$: $(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})$. f_X is Bernoulli$(\frac{1}{2})$.

Marginals do not determine the joint

f_1	$y=0$	$y=1$
$x=0$	$\frac{1}{4}$	$\frac{1}{4}$
$x=1$	$\frac{1}{4}$	$\frac{1}{4}$

f_2	$y=0$	$y=1$
$x=0$	$\frac{1}{2}$	0
$x=1$	0	$\frac{1}{2}$

Both have marginals $X \sim \text{Bernoulli}(\frac{1}{2})$, $Y \sim \text{Bernoulli}(\frac{1}{2})$, but they are different joints.

Continuous random vectors: joint pdf

Definition

(X, Y) is **continuous** with **joint pdf** $f \geq 0$ if for every region $A \subset \mathbb{R}^2$

$$P\big((X, Y) \in A\big) = \iint_A f(x, y)\, dx\, dy, \quad \iint_{\mathbb{R}^2} f = 1.$$

Marginal pdf

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy,$$
$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx.$$

Integrate out the variable you do not want. *Mind the support!*

Picture

$P(a < X < b) = \int_a^b f_X(x)\, dx$ is the volume above the strip $a < x < b$ of the plane. The marginal is that volume "squashed" onto the x-axis.

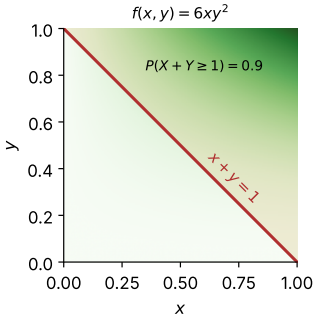
Example: $f(x, y) = 6xy^2$ on the unit square

$f(x, y) = 6xy^2,\ 0 < x < 1,\ 0 < y < 1$

- **Valid pdf:** $\int_0^1\int_0^1 6xy^2\,dy\,dx = 6 \cdot \frac{1}{2} \cdot \frac{1}{3} = 1.$
- **Marginals:** $f_X(x) = \int_0^1 6xy^2 dy = 2x,$
  $f_Y(y) = \int_0^1 6xy^2 dx = 3y^2.$
- **A probability:**

$$P(X + Y \geq 1) = 1 - \int_0^1 \int_0^{1-x} 6xy^2\,dy\,dx$$
$$= 1 - \int_0^1 2x(1-x)^3 dx = \tfrac{9}{10}.$$

- Note $f(x, y) = 2x \cdot 3y^2 = f_X(x) f_Y(y)$.



Joint cdf

Definition

$$F(x,y) = P(X \le x,\ Y \le y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f(u,v)\,dv\,du.$$

Properties

- $f(x,y) = \dfrac{\partial^2 F(x,y)}{\partial x\,\partial y}$ at continuity points.
- Marginal cdfs: $F_X(x) = \lim_{y\to\infty} F(x,y)$.
- $P(a < X \le b,\ c < Y \le d) = F(b,d) - F(a,d) - F(b,c) + F(a,c)$.

For $f = 6xy^2$ on the unit square

For $0 < x, y < 1$: $F(x,y) = \int_0^x 2u\,du \cdot \int_0^y 3v^2 dv = x^2y^3$.

Check: $\partial_x\partial_y\, x^2y^3 = 2x \cdot 3y^2 = 6xy^2$. $F_X(x) = x^2$, $F_Y(y) = y^3$.

Example: a non-rectangular support

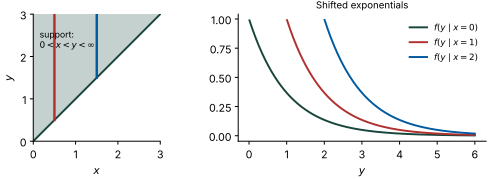
$f(x,y) = e^{-y}$ on $0 < x < y < \infty$

Valid: $\int_0^\infty \int_0^y e^{-y}\,dx\,dy = \int_0^\infty y e^{-y}\,dy = 1$.

- $f_X(x) = \int_x^\infty e^{-y}\,dy = e^{-x}$, $x > 0$: Exponential(1).
- $f_Y(y) = \int_0^y e^{-y}\,dx = y e^{-y}$, $y > 0$: Gamma(2, 1).

Limits of integration

The *region* dictates the limits: for f_X integrate y from x to ∞; for f_Y integrate x from 0 to y.



Left: support. Right: conditional densities (next section).

Example: uniform on a disk

$f(x,y) = \frac{1}{\pi}$ on $x^2 + y^2 < 1$

Marginal: for $|x| < 1$, y ranges over $|y| < \sqrt{1-x^2}$:

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi}\, dy = \frac{2}{\pi}\sqrt{1-x^2}, \quad |x| < 1.$$

A semicircle (Wigner) density, *not* uniform, although the joint is uniform.

Take-aways

- A joint that is flat can have non-flat marginals.
- $P(X^2 + Y^2 < \frac{1}{4}) = \frac{\pi/4}{\pi} = \frac{1}{4}$ is just **area ratio**.
- Here $f(x,y) \neq f_X(x) f_Y(y)$: X and Y are *not independent* (see §4.2).

Activity 1 — Joint and marginal

Part A: discrete

$f(x,y)$	$y=1$	$y=2$	$y=3$
$x=1$	0.10	0.15	0.15
$x=2$	0.20	0.25	0.15

(a) Find f_X and f_Y.

(b) $P(X+Y \le 3)$.

(c) $\mathbb{E}(XY)$, $\mathbb{E}X$, $\mathbb{E}Y$.

(d) Are X, Y independent? (compare $f(1,1)$ with $f_X(1)f_Y(1)$)

Part B: continuous

$f(x,y) = x+y$ on $0<x<1$, $0<y<1$.

(a) Verify that f is a pdf.

(b) Find the marginal f_X and $\mathbb{E}X$.

(c) Find $P(X+Y \le 1)$.

(d) Find $P(X<Y)$ (use symmetry).

Activity 1 — Solutions

Part A.

(a) $f_X = (0.40,\ 0.60)$ for $x = 1, 2$; $f_Y = (0.30,\ 0.40,\ 0.30)$ for $y = 1, 2, 3$.

(b) $(1,1), (1,2), (2,1)$: $0.10 + 0.15 + 0.20 = 0.45$.

(c) $\mathbb{E}X = 1.6$, $\mathbb{E}Y = 0.3 + 0.8 + 0.9 = 2.0$.
$\mathbb{E}(XY) = 1(0.10 + 0.30 + 0.45) + 2(0.20 + 0.50 + 0.45) = 0.85 + 2.30 = 3.15$ $(\neq \mathbb{E}X\mathbb{E}Y = 3.2)$.

(d) $f(1,1) = 0.10 \neq 0.40 \cdot 0.30 = 0.12$: **not independent**.

Part B.

(a) $\int_0^1 \int_0^1 (x + y)dy\, dx = \int_0^1 (x + \frac{1}{2})dx = 1$.

(b) $f_X(x) = x + \frac{1}{2}$, $0 < x < 1$; $\mathbb{E}X = \int_0^1 x(x + \frac{1}{2})dx = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$.

(c) $\int_0^1 \left[x(1-x) + \frac{(1-x)^2}{2}\right] dx = \int_0^1 \left(\frac{1}{2} - \frac{x^2}{2}\right) dx = \frac{1}{3}$.

(d) f is symmetric in (x, y), so $P(X < Y) = P(X > Y) = \frac{1}{2}$.

§4.2 Conditional distributions: motivation

Why condition?

- "Given that the first toss is heads, how many heads in total?"
- "Given the component lasted 2 years, how much longer will it last?"
- "Given today's height of the child, predict adult height."

We update the distribution of Y using the *observed* value of X: the foundation of regression and prediction.

From Chapter 1

For events, $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)}$. The conditional pmf/pdf is exactly this with $A = \{X = x\}$ and $B = \{Y = y\}$.

Careful with continuous X

For continuous X, $P(X = x) = 0$, so the ratio is $\frac{0}{0}$. The definition below fixes this by using *densities*.

Conditional pmf and pdf

Definition

Let (X, Y) have joint pmf/pdf $f(x, y)$ and marginal f_X. For any x with $f_X(x) > 0$ the **conditional pmf/pdf of Y given $X = x$** is

$$f(y \mid x) = \frac{f(x, y)}{f_X(x)}.$$

Similarly $f(x \mid y) = f(x, y)/f_Y(y)$ when $f_Y(y) > 0$.

It is a genuine distribution

$f(y \mid x) \geq 0$ and $\int f(y \mid x) dy = \frac{f_X(x)}{f_X(x)} = 1$. Probabilities:

$$P(Y \in A \mid X = x) = \int_A f(y \mid x)\, dy.$$

Product rule

Rearranging: $f(x, y) = f(y \mid x)\, f_X(x)$. A joint can be built from a marginal and a conditional (hierarchical models, §4.4).

Example: conditionals in the coin table

$Y \mid X = 1$

$f(y \mid X{=}1) = \dfrac{f(1,y)}{f_X(1)}$, with $f_X(1) = \frac{1}{2}$:

y	0	1	2	3
$f(y \mid 1)$	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

This is $1 + \mathrm{Binomial}(2, \frac{1}{2})$: first toss is H, then two free tosses.

$X \mid Y = 2$

$$P(X = 1 \mid Y = 2) = \frac{f(1,2)}{f_Y(2)} = \frac{2/8}{3/8} = \frac{2}{3}.$$

Given exactly two heads in three tosses, the first toss is heads with probability $\frac{2}{3}$ (two of the three positions of the tail are "after the first toss").

Example: conditional pdfs for $f(x, y) = e^{-y}$, $0 < x < y$

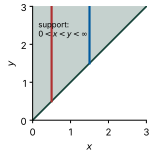
Using $f_X(x) = e^{-x}$, $f_Y(y) = ye^{-y}$

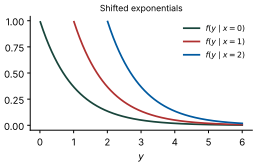
$$f(y \mid x) = \frac{e^{-y}}{e^{-x}} = e^{-(y-x)}, \quad y > x \quad \text{(shifted Exp(1))}$$

$$f(x \mid y) = \frac{e^{-y}}{ye^{-y}} = \frac{1}{y}, \quad 0 < x < y \quad \text{(Uniform}(0, y)\text{)}$$

A conditional probability

$P(Y > 3 \mid X = 1) = \int_3^\infty e^{-(y-1)} dy = e^{-2} \approx 0.135.$





Interpretation

Given $X = x$, the excess $Y - x$ is Exp(1) regardless of x: a memoryless structure.

Conditional expectation and variance

Definition

$$\mathbb{E}(g(Y)\,|\,x) = \sum_y g(y) f(y\,|\,x) \quad \Big(\text{or } \int g(y) f(y\,|\,x)\,dy\Big),$$

$$\mathbb{E}(Y\,|\,x), \qquad \mathrm{Var}(Y\,|\,x) = \mathbb{E}(Y^2\,|\,x) - \big[\mathbb{E}(Y\,|\,x)\big]^2.$$

Both are *functions of* x; $\mathbb{E}(Y\,|\,X)$ is a random variable.

Back to $f(x,y) = e^{-y}$, $0 < x < y$

$Y\,|\,x \sim x + \mathrm{Exp}(1)$:
$\mathbb{E}(Y\,|\,x) = x + 1$, $\mathrm{Var}(Y\,|\,x) = 1$.
$X\,|\,y \sim \mathrm{Uniform}(0, y)$:
$\mathbb{E}(X\,|\,y) = \frac{y}{2}$, $\mathrm{Var}(X\,|\,y) = \frac{y^2}{12}$.

Preview (§4.4)

$\mathbb{E}Y = \mathbb{E}\big[\mathbb{E}(Y\,|\,X)\big] = \mathbb{E}(X+1) = 1 + 1 = 2$: matches Gamma$(2,1)$ with mean 2. Averaging the conditional mean recovers the unconditional mean.

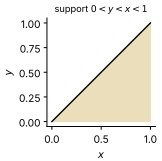
Example: a broken stick

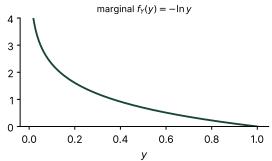
Build a joint from a conditional

Pick $X \sim \mathrm{Uniform}(0,1)$, then $Y \mid X = x \sim \mathrm{Uniform}(0,x)$ (break a stick, then break the piece again).

$$f(x,y) = f(y \mid x) f_X(x) = \frac{1}{x} \cdot 1, \quad 0 < y < x < 1.$$

$$f_Y(y) = \int_y^1 \frac{1}{x} dx = -\ln y, \quad 0 < y < 1.$$





- $P(Y > \frac{1}{2}) = \int_{1/2}^1 (-\ln y) dy = \frac{1}{2} - \frac{1}{2}\ln 2 \approx 0.153.$
- $\mathbb{E}Y = \mathbb{E}\left[\frac{X}{2}\right] = \frac{1}{4} = \int_0^1 y(-\ln y) dy.$

Independence

Definition

Continuous or discrete (X, Y) with joint $f(x, y)$ and marginals f_X, f_Y are **independent** if for *all* x, y

$$f(x, y) = f_X(x)\, f_Y(y).$$

Equivalent statements

- $f(y \mid x) = f_Y(y)$ for all x with $f_X(x) > 0$;
- $f(x \mid y) = f_X(x)$;
- $P(X \in A,\ Y \in B) = P(X \in A)P(Y \in B)$ for all sets A, B.

Meaning

Learning X tells you **nothing** about Y: the conditional distribution of Y does not change with x. Independence is a property of the *whole* joint distribution, not of one cell.

A practical test: the factorization lemma

Lemma (factorization)

X and Y are independent if and only if there exist functions $g(x)$ and $h(y)$ such that for all x, y

$$f(x, y) = g(x)\, h(y).$$

(No need to compute the marginals: g, h need not be pdfs, only proportional to them.)

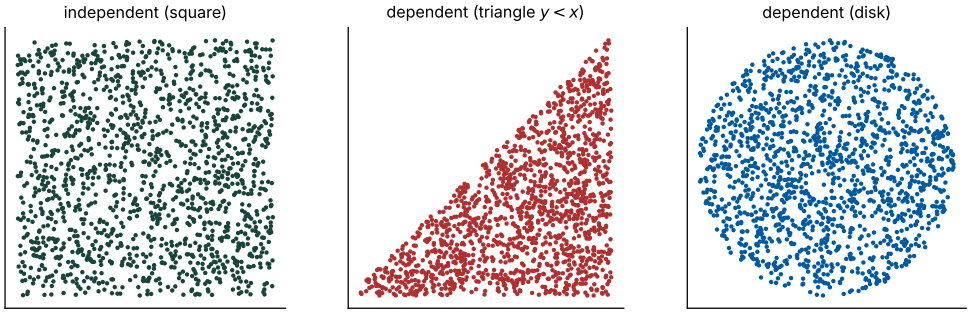
Independent?

- $f = 6xy^2$ on the unit square: $g = 6x$, $h = y^2$: **yes**.
- $f = e^{-x-y}$, $x, y > 0$: $g = e^{-x}$, $h = e^{-y}$: **yes** (iid Exp(1)).
- $f = x + y$ on the unit square: cannot factor: **no**.

Watch the support!

$f = e^{-y}$ on $0 < x < y$ *looks* like $g(x)h(y)$ with $g = 1$, $h = e^{-y}$, but the indicator $I(x < y)$ couples x and y: **not independent**. The support must be a *rectangle* (a product set) for independence.

Pictures of independence



Visual cue

If (X, Y) is independent, the conditional spread of Y is the same at every x. A triangle or a disk fails: the vertical range of Y depends on x.

Careful

A rectangular support is *necessary* but not sufficient: $f = x + y$ on the unit square has a rectangular support but is not a product.

Expectations under independence

Theorem

Let X and Y be independent. Then

(a) for any functions g, h: $\mathbb{E}\big[g(X)h(Y)\big] = \mathbb{E}g(X) \cdot \mathbb{E}h(Y)$;

(b) the mgf of the sum is $M_{X+Y}(t) = M_X(t)\, M_Y(t)$.

Proof

(a) $\mathbb{E}[g(X)h(Y)] = \int\int g(x)h(y)f_X(x)f_Y(y)\,dx\,dy = \int gf_X \cdot \int hf_Y$.
(b) Take $g(x) = e^{tx}$, $h(y) = e^{ty}$ in (a): $M_{X+Y}(t) = \mathbb{E}[e^{tX}e^{tY}] = M_X(t)M_Y(t)$.

Converse is false

$\mathbb{E}(XY) = \mathbb{E}X\mathbb{E}Y$ does **not** imply independence. (Covariance and uncorrelated variables: §4.5.) E.g. $X \sim U(-1,1)$ and $Y = X^2$: $\mathbb{E}(XY) = \mathbb{E}X^3 = 0 = \mathbb{E}X\mathbb{E}Y$, yet Y is a function of X.

Sums of independent variables via mgfs

Poisson

$X \sim \text{Poisson}(\theta)$, $Y \sim \text{Poisson}(\lambda)$ independent.

$$M_{X+Y}(t) = e^{\theta(e^t-1)} e^{\lambda(e^t-1)} = e^{(\theta+\lambda)(e^t-1)}.$$

So $X + Y \sim \text{Poisson}(\theta + \lambda)$.

Gamma, same scale

$\text{Gamma}(\alpha_1, \beta) + \text{Gamma}(\alpha_2, \beta) = \text{Gamma}(\alpha_1 + \alpha_2, \beta)$ since $(1 - \beta t)^{-\alpha_1}(1 - \beta t)^{-\alpha_2} = (1 - \beta t)^{-(\alpha_1+\alpha_2)}$.

Sum of independent normals

$X \sim N(\mu, \sigma^2)$, $Y \sim N(\gamma, \tau^2)$ independent:

$$M_{X+Y}(t) = e^{(\mu+\gamma)t+(\sigma^2+\tau^2)t^2/2},$$

i.e. $X + Y \sim N(\mu + \gamma,\ \sigma^2 + \tau^2)$.

Numerical

$X \sim N(1, 4)$, $Y \sim N(3, 5)$ independent. Then $X + Y \sim N(4, 9)$ and

$$P(X + Y > 6) = P\left(Z > \tfrac{2}{3}\right) \approx 0.2525.$$

Functions of independent variables; a probability by double integral

Functions preserve independence

If X and Y are independent, then $g(X)$ and $h(Y)$ are independent for any functions g, h.

Example

X, Y iid $N(0,1)$ are independent, hence so are X^2 and Y^2, and $X^2 + Y^2 \sim \chi^2_2$ (sum of independent gammas).

Race between two lifetimes

$X \sim \text{Exp}$ (mean 2) and $Y \sim \text{Exp}$ (mean 3), independent. Find $P(X < Y)$.

$$P(X < Y) = \int_0^\infty \int_x^\infty \tfrac{1}{2} e^{-x/2} \tfrac{1}{3} e^{-y/3} dy\, dx$$
$$= \int_0^\infty \tfrac{1}{2} e^{-5x/6} dx = \frac{1/2}{5/6} = \tfrac{3}{5}.$$

In general $P(X < Y) = \frac{\lambda_X}{\lambda_X + \lambda_Y}$.

Activity 2 — Conditionals and independence

Problems

Let (X, Y) be uniform on the triangle $0 < y < x < 1$, that is $f(x, y) = 2$ there.

(a) Find the marginals $f_X(x)$ and $f_Y(y)$.

(b) Find $f(y \mid x)$ and name the conditional distribution. Find $\mathbb{E}(Y \mid X = x)$.

(c) Find $P(Y > \frac{1}{2} \mid X = \frac{3}{4})$.

(d) Are X and Y independent? Give two reasons.

Independent pair. $X \sim \text{Poisson}(2)$ and $Y \sim \text{Poisson}(3)$ are independent.

(e) Name the distribution of $X + Y$ and find $P(X + Y = 0)$.

(f) Find $\mathbb{E}(XY)$.

Activity 2 — Solutions

(a) $f_X(x) = \int_0^x 2\,dy = 2x$ $(0 < x < 1)$; $f_Y(y) = \int_y^1 2\,dx = 2(1-y)$ $(0 < y < 1)$.

(b) $f(y \mid x) = \dfrac{2}{2x} = \dfrac{1}{x}$, $0 < y < x$: Uniform$(0, x)$; $\mathbb{E}(Y \mid x) = \frac{x}{2}$.

(c) Uniform$(0, \frac{3}{4})$: $P(Y > \frac{1}{2}) = \dfrac{3/4 - 1/2}{3/4} = \frac{1}{3}$.

(d) **No.** (i) The support is a triangle, not a rectangle. (ii) $f(y \mid x) = \frac{1}{x}$ depends on x (equivalently $f(x, y) = 2 \neq 4x(1-y) = f_X f_Y$).

(e) $X + Y \sim \mathrm{Poisson}(5)$ (mgf product); $P(X + Y = 0) = e^{-5} \approx 0.0067$.

(f) Independence: $\mathbb{E}(XY) = \mathbb{E}X\,\mathbb{E}Y = 2 \cdot 3 = 6$.

Summary: discrete vs. continuous

	Discrete	**Continuous**
Joint	$f(x,y) = P(X = x, Y = y)$	$P((X,Y) \in A) = \int\int_A f$
Marginal	$f_X(x) = \sum_y f(x,y)$	$f_X(x) = \int f(x,y)dy$
Conditional	$f(y \mid x) = \dfrac{f(x,y)}{f_X(x)}$	$f(y \mid x) = \dfrac{f(x,y)}{f_X(x)}$
Independence	$f(x,y) = f_X(x)f_Y(y)$	$f(x,y) = f_X(x)f_Y(y)$
$\mathbb{E}g(X,Y)$	$\sum\sum g\, f$	$\int\int g\, f$

Key ideas

Marginals come from the joint by summing/integrating; the joint does *not* come from the marginals unless the variables are independent. Conditionals rescale a slice of the joint. Independence $\Leftrightarrow$ the joint factors (on a product support).

Wrap-up and next time

Check yourself

- Can you get marginals from a table and from a pdf (with the right limits)?
- Can you form $f(y \mid x)$ and compute $\mathbb{E}(Y \mid x)$?
- Can you test independence by factorization (and check the support)?
- Do you know why independence gives $\mathbb{E}[g(X)h(Y)] = \mathbb{E}g(X)\mathbb{E}h(Y)$ and $M_{X+Y} = M_X M_Y$?

Coming up

- §4.3 Bivariate transformations and the Jacobian
- §4.4 Hierarchical models, mixtures, $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X \mid Y)]$
- §4.5 Covariance and correlation